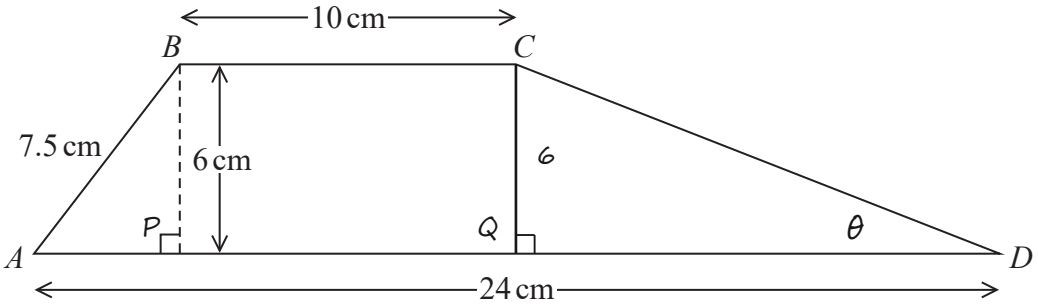
7 *ABCD* is a trapezium.



Work out the size of angle *CDA*.
Give your answer correct to 1 decimal place.

P is below B and Q is below C on AD.

In right-angled triangle ABP: $AP^2 + 6^2 = 7.5^2$.

1 mark
Use Pythagoras to find the left base

$AP = \sqrt{(7.5^2 - 6^2)} = \sqrt{20.25} = 4.5$ cm

1 mark
Complete the calculation of AP

PQ = BC = 10 cm, so QD = 24 − 10 − 4.5 = 9.5 cm.

1 mark
Find the base of the right-hand triangle

For θ = angle CDA: opposite = 6, adjacent = 9.5.

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{6}{9.5}$$

1 mark
Use the tangent ratio

$\theta = \tan^{-1}(6 \div 9.5) = 32.2756...°$

To 1 decimal place: angle CDA = 32.3°

1 mark
Correct angle; accepted 32.2°–32.3°

32.3 °

(Total for Question 7 is 5 marks)